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22IT801 – Project - II

Intrusion Detection System using Voting-Based Neural Network

DOMAIN: CYBERSECURITY & MACHINE LEARNING

Intrusion Detection System Using Voting-Based Neural Network

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ABSTRACT

- Modern IDS face difficulties in handling high-speed and complex network traffic, often causing missed intrusions or generating excessive false alerts that reduce system reliability.
- The proposed system uses multiple Deep Neural Network models trained on different traffic patterns to capture diverse and hidden attack behaviours effectively.
- A voting-based decision mechanism combines model outputs to improve detection accuracy, reduce alert fatigue, and strengthen protection against zero-day attacks.

INTRODUCTION

- As cyberattacks grow more complex, traditional security systems often struggle with high false alarm rates and missed threats.
- This project develops an Intrusion Detection System (IDS) powered by a Voting-Based Neural Network architecture.
- By combining the predictions of multiple diverse deep learning models, the system acts like a "digital jury" to ensure high-confidence decisions.
- This ensemble approach effectively captures complex traffic patterns that single models might miss, significantly reducing false positives while increasing resilience against new, unseen attacks.
- The result is a more reliable and automated defense framework capable of securing modern, high-speed network infrastructures.

PROBLEM DEFINITION

- Modern high-speed networks and the rapidly evolving threat landscape have created a critical gap in network security.
- Traditional Intrusion Detection Systems (IDS) are failing to maintain satisfactory performance due to their reliance on static attack signatures and brittle classification models.
- The core challenge is the lack of an intelligent, resilient security framework capable of simultaneously achieving near-perfect detection of novel, sophisticated cyberattacks (low false negatives) while drastically minimizing misclassification of normal traffic (low false positives).
- This failure results in unmanageable alert volumes, security staff burnout, and critical vulnerability to advanced threats

OBJECTIVE

- Build Multiple Neural Networks: Develop three different neural network models — Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM) — to analyze and classify network traffic data effectively.
- Combine Decisions Using Voting: Develop a simple voting system (like "majority rules") to take the final prediction from all the built networks and output a single, final security decision (Attack or Normal).
- Achieve High Accuracy: Show that the combined Voting system gets a significantly higher overall correct detection rate (Accuracy) than any of the single networks alone.
- Reduce False Alarms: Demonstrate that the voting system minimises the number of incorrect "Attack" warnings (False Positives) compared to the individual base networks.
- The system was trained and evaluated using a Kaggle IDS dataset and a manually generated dataset, achieving high detection accuracy through a voting ensemble of MLP, CNN, and LSTM models.

LITERATURE SURVEY

Sl.No.	Paper published Year, Vol. Pages	Authors' Name	Paper Title & Journal Name	Findings
[1]	20 February 2025, Vol. 13	Methaq A. Shyaadi, Noor Farizah Ibrahim, Zurinahni Binti Zainol, Rosni Abdullah, and Mohammed Anbar.	Reinforcement Learning-Based Voting for Feature Drift-Aware Intrusion Detection: An Incremental Learning Framework.	The researchers developed the Incremental Feature Drift-Aware Genetic Programming Combiner (IFDA-GPC), which utilizes a multi-agent reinforcement learning voting mechanism to achieve superior detection performance across multiple benchmark datasets, including an accuracy of 88% on the CICIDS-2017 dataset.
[2]	Vol. 15, No. 3, 2025, 23917-23922	Kumar, R., Singh, P., & Patel, A.	Development of an Intrusion Detection System using an Ensemble Voting Machine Learning Technique.	The study developed an ensemble voting classifier combining Naïve Bayes (NB), K-Nearest Neighbors (KNN), and Adaptive Boosting (AdaBoost) that achieved a superior accuracy of 89.79% on the KDD99 dataset, outperforming individual classifiers.
[3]	Vol. 09 No. 9, August 2025.	Araoluwa Simileolu Filani, Olasupo Modupe Adegoke, Abimbola Abosede Joseph, and Odewale Abdullahi Opeyemi.	Development of a Fake Job Posting Detection System using Deep Neural Networks and Voting Ensemble Methods.	The "Ensemble DNN with Voting" model achieved superior performance in distinguishing between real and fraudulent job listings, specifically reaching 86% accuracy, 84% precision, 86% recall, and an Area Under the Curve (AUC) of 0.88
[4]	October 2025, Computers & Security, Volume 157.	Syed Wali.Yasir Ali Farrukh.Irfan Khan.	Explainable AI and Random Forest based reliable intrusion detection system.	The research introduces a novel two-stage IDS framework that achieves high resilience against advanced black-box adversarial attacks without requiring expensive retraining by combining SHAP-driven credibility assessments with a secondary pipeline featuring Transformer-based semantic payload inspection

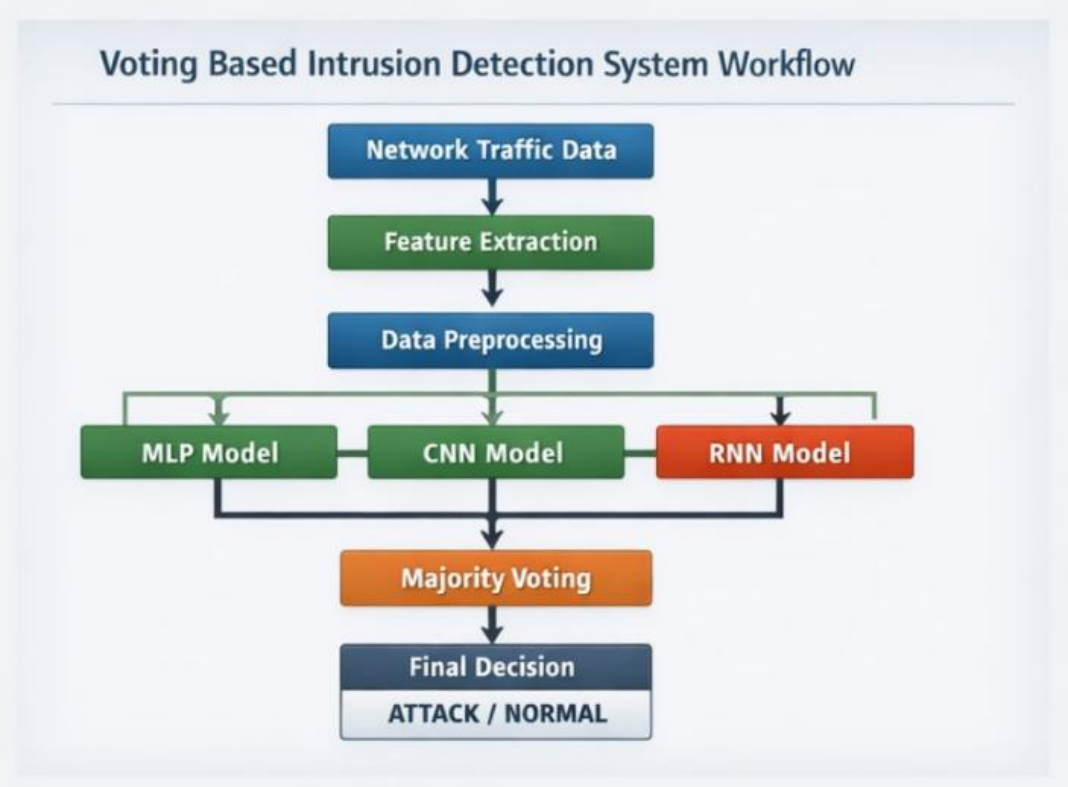
Sl.No.	Paper published Year, Vol. Pages	Authors' Name	Paper Title & Journal Name	Findings
[5]	Volume 10, Issue 4, April 2025.	Krotha Karunya, Lingireddy Anuradha, Katta Ravindra, and Sri. P. Rama Krishna.	A Voting Based Ensemble Approach for Fake Account Detection using Multimodal Social Media Data.	The research developed a voting-based ensemble model that achieved 98.01% accuracy and a 98.10% F1-score by integrating XGBoost, Random Forest, and Gaussian Naive Bayes, significantly outperforming individual deep learning models like CNN, LSTM, and ANN in detecting fraudulent social media accounts.
[6]	Sept 2025, version 2509.10206v1.	Federica Uccello and Simin Nadjm-Tehrani.	Investigating Feature Attribution for 5G Network Intrusion Detection.	The research demonstrated that the logic-based VOTE-XAI method provides more concise, stable, and computationally efficient explanations for 5G security alerts than the statistical SHAP method, generating a single explanation in under 0.002 seconds even in high-dimensional settings with 478 features
[7]	2025, Scientific Reports, Vol. 15, Article No. 14313	Yousra Berrich	EEG-based epilepsy detection using CNN-SVM and DNN-SVM with feature dimensionality reduction by PCA	The study demonstrates that integrating Principal Component Analysis (PCA) with hybrid deep learning models (CNN-SVM and DNN-SVM) significantly enhances the accuracy of automatic epilepsy detection, achieving a peak accuracy of 89.96% on the BONN dataset
[8]	2025, Electronics, Vol. 14, No. 1176.	Mark Devine , Saeid Pourroostaei Ardakani Mohammed Al-Khafajiy Yvonne James	Federated Machine Learning to Enable Intrusion Detection Systems in IoT Networks.	The study introduces a first-of-its-kind federated Support Vector Machine (SVM) model for IoT intrusion detection that effectively balances high attack prediction accuracy with the low computational and memory requirements necessary for deployment on resource-constrained edge devices.

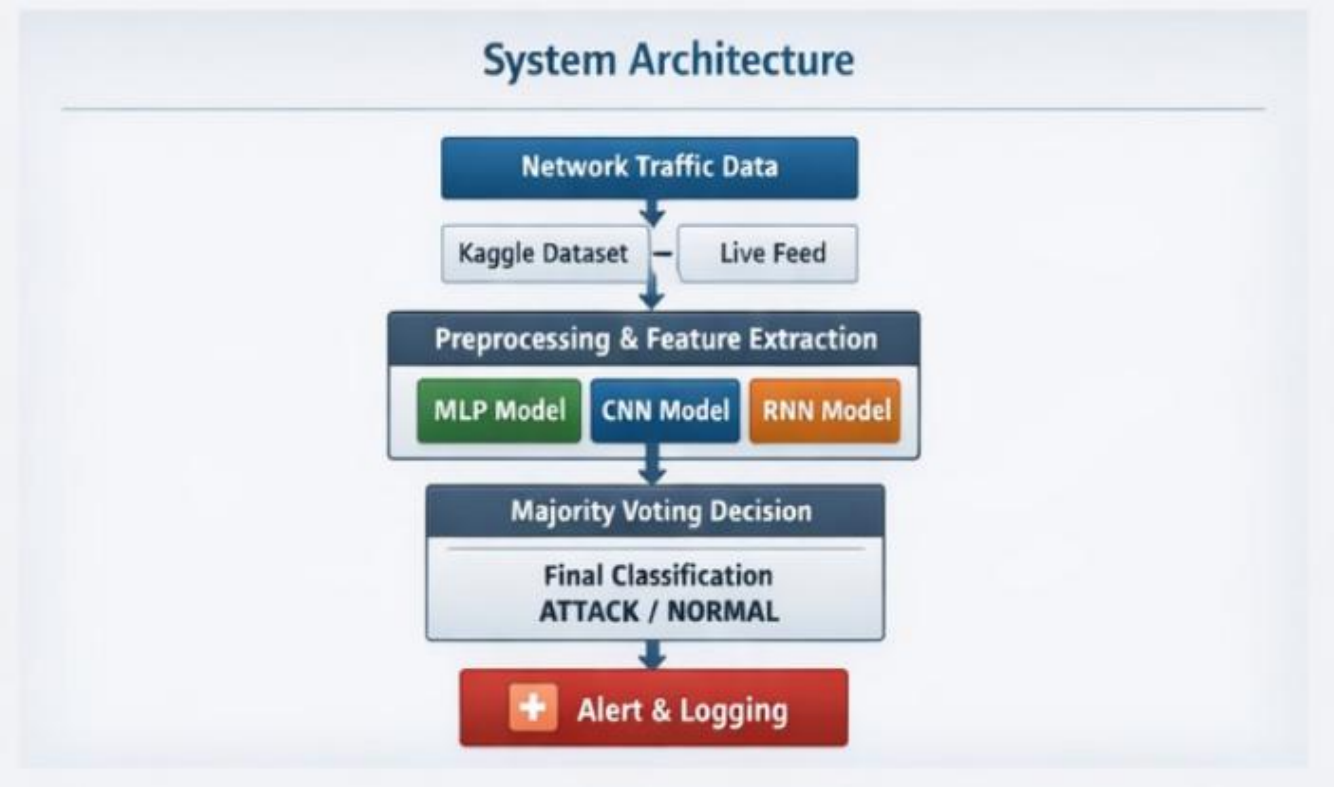
Sl.No.	Paper published Year, Vol. Pages	Authors' Name	Paper Title & Journal Name	Findings
[9]	2025, Cluster Computing, Vol. 28, Article No. 473	Busra Buyuktanir Şahsene Altinkaya Gozde Karatas Baydogmus Kazim Yildiz	Federated learning in intrusion detection: advancements, applications, and future directions	This research demonstrates that incorporating Federated Learning (FL) into Intrusion Detection Systems (IDS) effectively addresses data privacy and security challenges by allowing collaborative model training across decentralized systems without centralizing sensitive data
[10]	2025, Artificial Intelligence Review, Vol. 58, Article No. 114	Arezou Naghib Farhad Soleimanian Gharehchopogh Azadeh Zamanifar	A comprehensive and systematic literature review on intrusion detection systems in the internet of medical things: current status, challenges, and opportunities	This systematic review identifies that while artificial intelligence models—particularly Random Forest, Decision Trees, and Long Short-Term Memory (LSTM) networks—are highly effective for securing Internet of Medical Things (IoMT) environments, there remains a critical need for lightweight, anomaly-based frameworks capable of real-time detection in resource-constrained medical devices
[11]	2025, Electronics, Vol. 14, No. 261	Jiachen Huang Yuling Chen Xuewei Wang Zhi Ouyang Nisuo Du	Optimization Scheme of Collaborative Intrusion Detection System Based on Blockchain Technology	The paper introduces the Blockchain-based Collaborative Intrusion Detection Framework (BCIDF), which incorporates a Weighted Random Forest (WRF) ensemble learning algorithm and an Alternating Random Assignment Selection Mechanism (ARASM) to significantly enhance detection precision and alert-sharing efficiency compared to traditional intrusion detection systems
[12]	2025, Scientific Reports, Vol. 15, Article No. 14177.	Zeinab Awad Magdy Zakaria Rasha Hassan	An enhanced ensemble defense framework for boosting adversarial robustness of intrusion detection systems.	The research demonstrates that an ensemble defense framework—combining adversarial training, label smoothing, and Gaussian augmentation during training with a denoising sparse autoencoder during testing—significantly improves the robustness of deep learning-based IDS against adversarial attacks, achieving prediction accuracies of up to 88.78%.

EXISTING SYSTEM

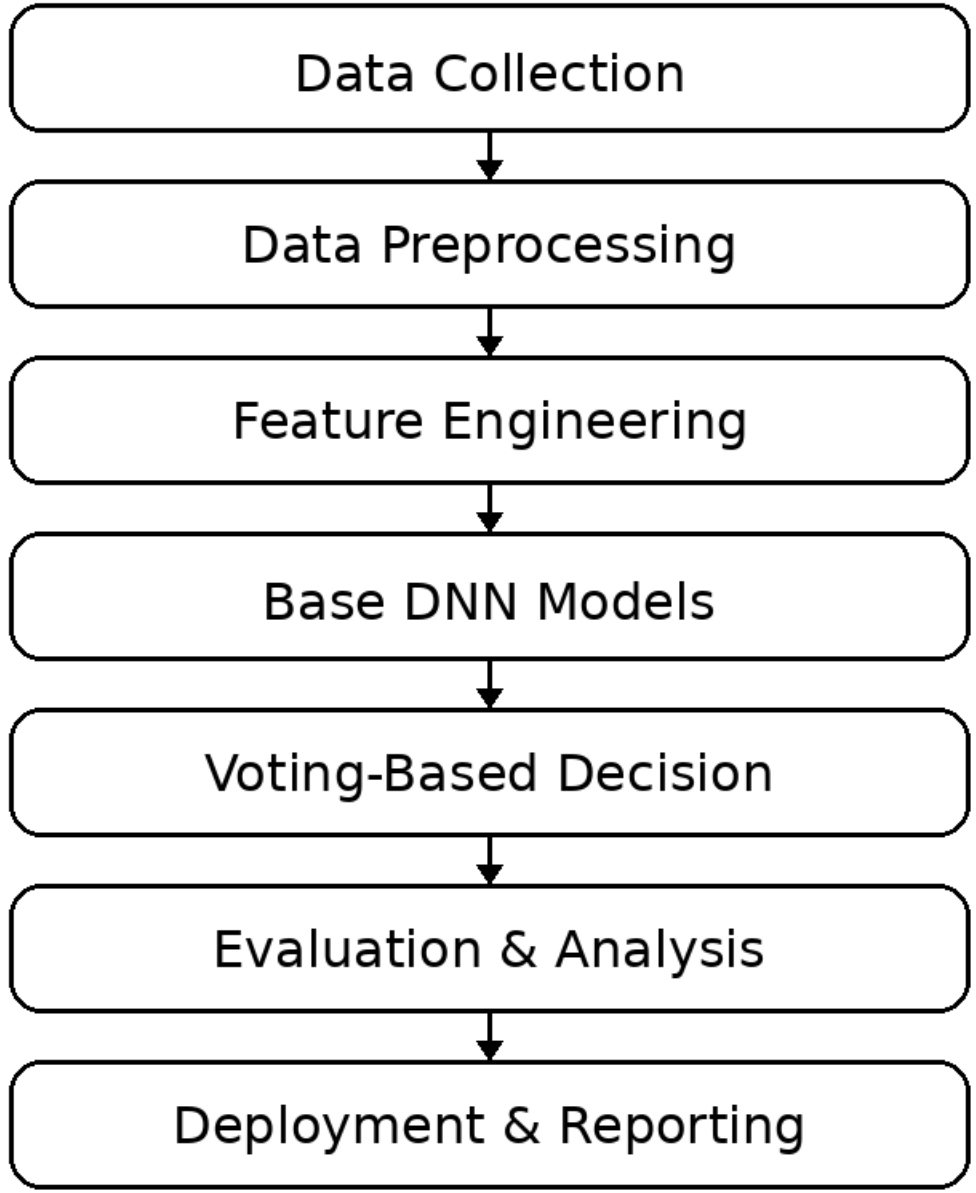
- Existing Intrusion Detection Systems (IDS) mainly use static, rule-based methods and fixed databases of known attack signatures.
- Current systems often rely on Genetic Programming Combiners (GPC) that link multiple classifiers but struggle with changing data.
- Many models are static, meaning they are trained once and cannot automatically update when new types of cyber threats appear.
- Traditional frameworks focus on Concept Drift (changes in attack patterns) but completely ignore Feature Drift (changes in which data points are relevant).
- Existing feature selection methods use random search techniques like Particle Swarm Optimization (PSO), which are slow and often inaccurate.
- Most current systems require frequent manual updates and human intervention to stay effective against modern hackers.

SYSTEM ARCHITECTURE

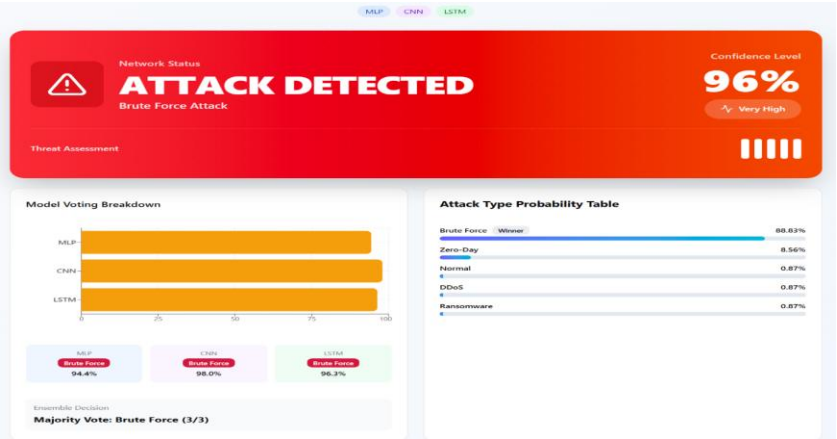


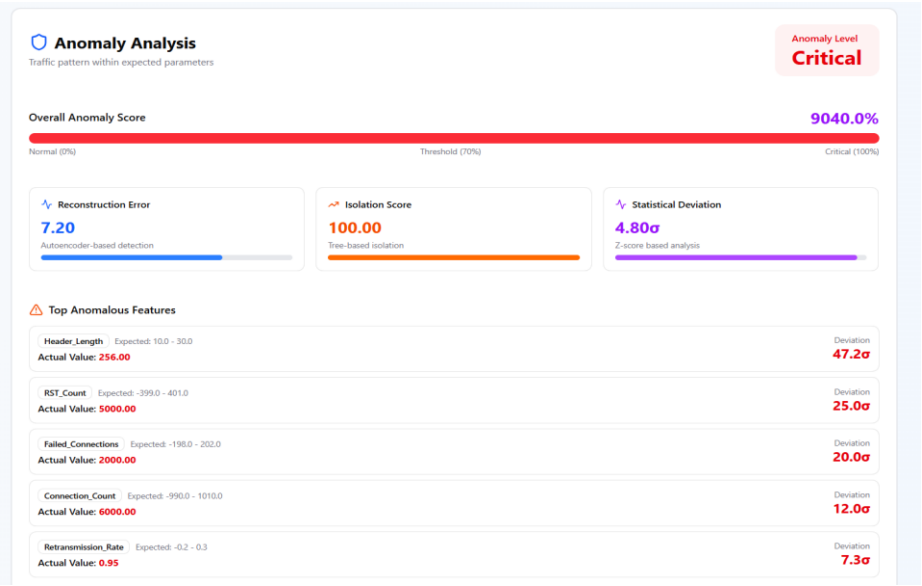


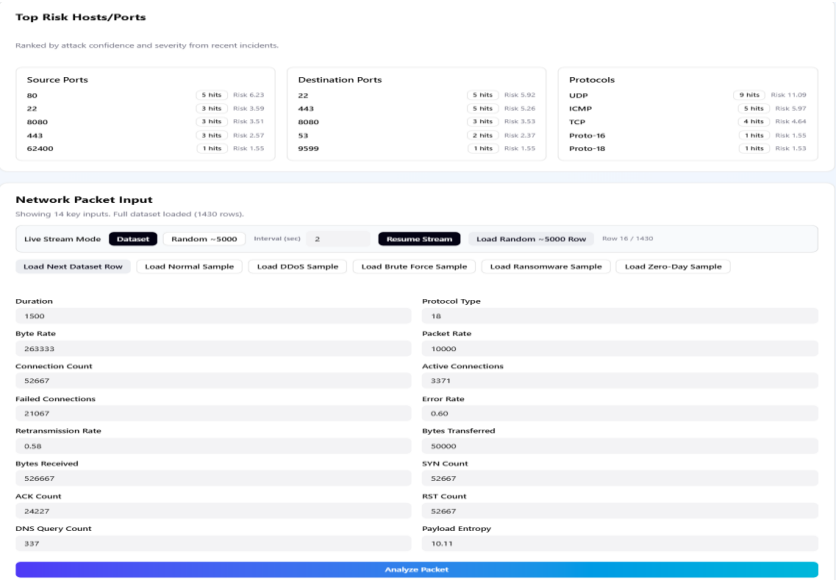
Module Design

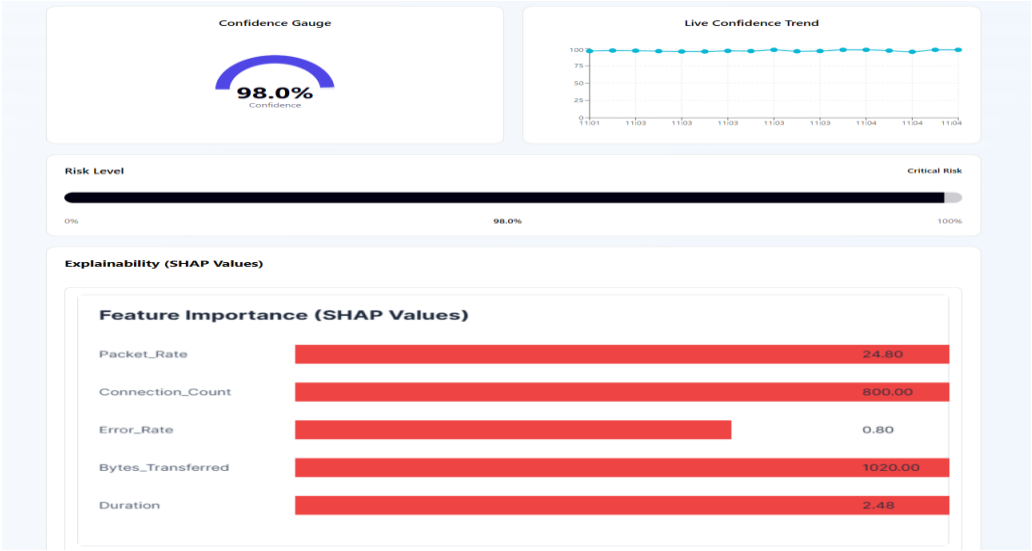


OUTPUT SCREENSHOTS





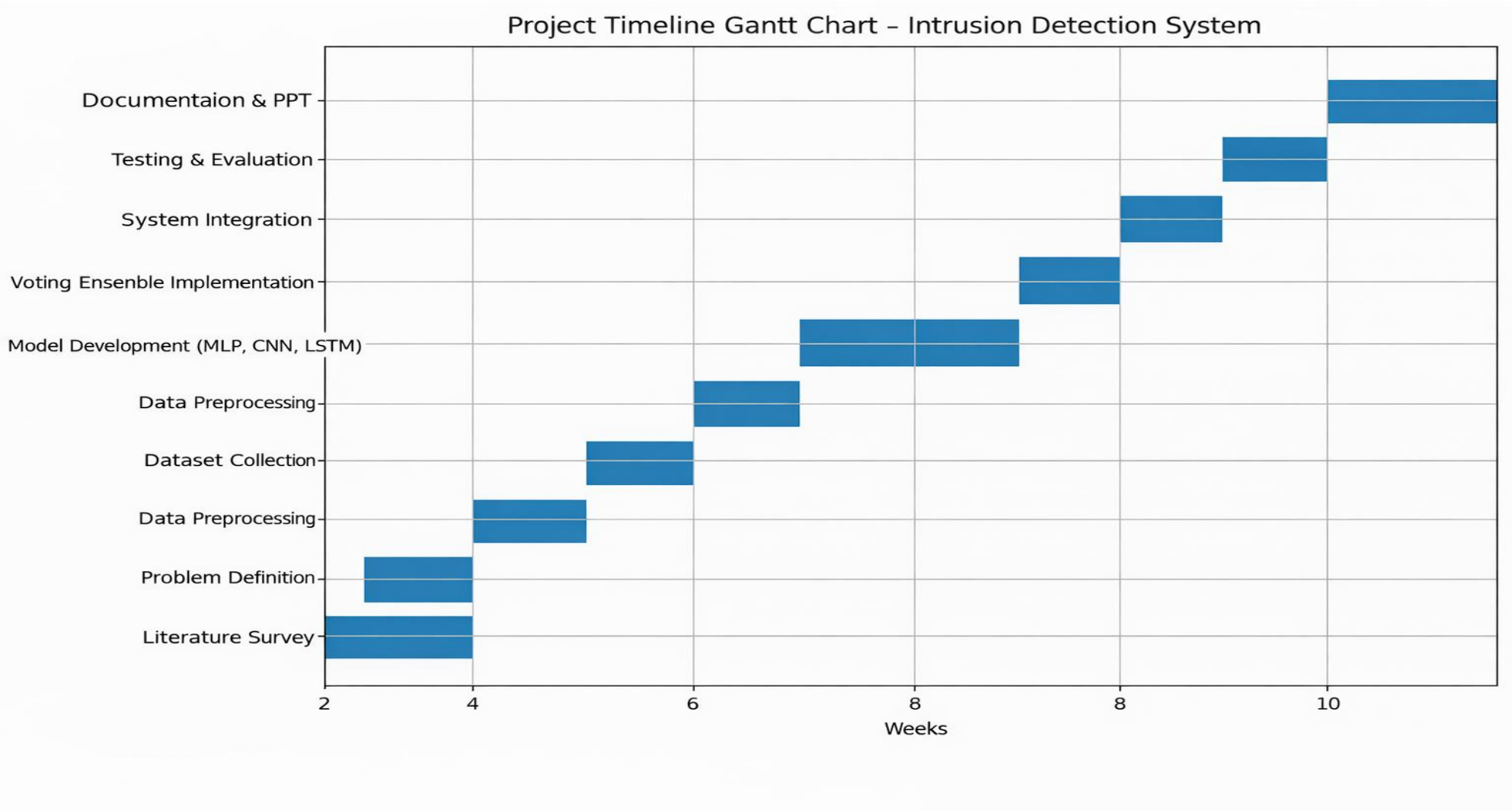




PROPOSED SYSTEM

- Higher Detection Accuracy using a voting-based ensemble of multiple neural networks.
- Reduced False Alarms through consensus decision-making among models.
- Zero-Day Attack Detection using behavioral anomaly analysis.
- Heartbeat Monitoring detects model failures and ensures system reliability.
- SHA-256 Security protects system data and model outputs from tampering.
- Parallel Processing enables efficient analysis of high-volume network traffic.
- Automatic Feature Learning reduces dependency on manual signatures.
- Reliable and Scalable System suitable for real-world cybersecurity environments.

TIME LINE


Project Timeline Gantt Chart – Intrusion Detection System
Documentaion & PPT
Testing & Evaluation
System Integration
Voting Ensenble Implementation
Model Development (MLP, CNN, LSTM)
Data Preprocessing
Dataset Collection
Data Preprocessing
Problem Definition
Literature Survey
2
4
6
8
8
10
Weeks

Technology Stack

Front End

React TypeScript Vite Tailwind CSS

Backend / Coding Language

python™

Framework

PyTorch

CONCLUSION

The proposed Voting-Based Neural Network framework effectively addresses the limitations of static and single-model defense systems. By combining multiple specialized neural networks through an intelligent voting process, the system achieves higher detection accuracy and a significant reduction in false alarms. To enhance system reliability, a heartbeat signal mechanism is incorporated to continuously monitor the operational status of each model, enabling rapid detection of model failures or inactive components within the framework. Additionally, to ensure data integrity and secure communication within the system, SHA-256 cryptographic hashing is implemented, protecting model outputs and system data from tampering or unauthorized modification. This approach ensures a resilient, secure, and adaptive security infrastructure capable of protecting networks against both known and evolving cyber threats.

REFERENCES

[1] Methaq A. Shyaa, Noor Farizah Ibrahim, Zurinahni Binti Zainol, Rosni Abdullah, Mohammed Anbar, "Reinforcement Learning-Based Voting for Feature Drift-Aware Intrusion Detection: An Incremental Learning Framework", in IEEE Access, https://doi.org/10.1109/ACCESS.2025.3544221, 2025.

[2] Yousra Berrich, Zouhair Guennoun, "EEG-based epilepsy detection using CNN-SVM and DNN-SVM with feature dimensionality reduction by PCA", in Scientific Reports, https://doi.org/10.1038/s41598-025-95831-z, 2025.

[3] Mark Devine, Saeid Pourroostaei Ardakani, Mohammed Al-Khafajiy, Yvonne James, "Federated Machine Learning to Enable Intrusion Detection Systems in IoT Networks", in Electronics, https://doi.org/10.3390/electronics14061176, 2025.

[4] Busra Buyuktanir, Şahsene Altinkaya, Gozde Karatas Baydogmus, Kazim Yildiz, "Federated learning in intrusion detection: advancements, applications, and future directions", in Cluster Computing, https://doi.org/10.1007/s10586-025-05325-w, 2025.

[5] Arezou Naghib, Farhad Soleimanian Gharehchopogh, Azadeh Zamanifar, "A comprehensive and systematic literature review on intrusion detection systems in the internet of medical things: current status, challenges, and opportunities", in Artificial Intelligence Review, https://doi.org/10.1007/s10462-024-11101-w, 2025.

CONFERENCE DETAILS AND SCREENSHOTS

CONFERENCE: International Conference on Connected Intelligence for Industrial Applications (CI2A-2026) at Akal University, Punjab, India is accepted for publication in IEEE Xplore proceedings (Scopus & Web of Science).

Microsoft CMT <noreply@msr-cmt.org> Tue, Feb 17, 8:03 PM

to me

Dear authors,
Congratulations for your paper acceptance & registration in CI2A 2026.

We are glad to let you know that our conference will be held during 3rd - 5th April, 2026.
In this context, for smooth proceedings of the event, we request you to join our conference's official Whatsapp group by tomorrow positively for further information about the event.

Whatsapp Group Link: https://chat.whatsapp.com/HbdyhYnHYCK2BbBWYR8Rhf?mode=gi_t

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